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Reyhaneh(Rey) Kharazmi Pour

Optimization & Operations management

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PhD student in Operations Management with a strong foundation in data science, optimization, and statistical analysis. Experienced in A/B testing, customer segmentation, and predictive modeling using Python and SQL. Passionate about applying ML and data-driven insights to real-world problems.

EDUCATION

Ph.D. in Operations, Decisions, and Technology Management, *University of California, Irvine*

GPA: 4.0

2023- Present

Teaching Assistant Experience: Management Science Analytics, Operations Management, Statistics for Business Decision Making, Analytical Decision Models for Managers

B.Sc. in Industrial Engineering, *Sharif University of Technology*

GPA of Last 2 years: 4.0

2017-2022

Thesis title: "Optimal referral marketing with simulation-optimization approach" - [Read here](#)

Teaching Assistant Experience: Project control, Transportation Planning, Facilities Planning, Time and Work Evaluating

Diploma in Mathematics and Physics Discipline,

Farzanegan High School, Affiliated with the National Organization for Development of Exceptional Talents (NODET)

2013-2017

RESEARCH EXPERIENCE

Learning to Monetize Influence: Online Ad Allocation with Edge-Aware Dual Mirror Descent

Coauthor: Professor John Turner

Ongoing

Developing online algorithms for display ad assignment in social networks, leveraging edge-aware dual mirror descent for near-optimal allocation under budget constraints.

Regulating Farmland Abandonment under Network Cost Spillovers

Coauthor: Professor Luyi Gui

Ongoing

This paper studies farmland abandonment in agricultural markets with network cost spillovers and analyzes how penalty-based and subsidy-based policies influence farmers' participation and production decisions.

Optimal Paid Referral Marketing

Coauthor: Professor Erfan HassanNayebi [Abstract](#)

2022

Designed an agent-based simulation and optimization framework to plan referral marketing campaigns that maximize user acquisition under a fixed budget.

EXPERIENCE

Business Data Analyst

Jul. 2021 — Aug. 2023

Digikala.com (The largest e-commerce business in Iran)

Tehran, IRAN

- Collaborated cross-functionally with marketing, product, sales, operations, and business development teams to support the launch and growth of Digikala's on-demand grocery and food delivery service
- Conducted funnel analysis and conversion rate optimization using Google Analytics and Hotjar; designed and evaluated A/B tests to improve user experience and checkout flow
- Analyzed P&L data and operational KPIs to inform strategic decisions on pricing, assortment, and promotions
- Scraped competitor websites to identify assortment gaps and provided actionable recommendations to improve product coverage
- Performed customer segmentation and lifecycle analysis, including churn prediction and CLV modeling, to support personalized marketing and retention strategies
- Supported ECRM (Email & Customer Relationship Management) journey design by analyzing customer behavior and campaign effectiveness
- Investigated operational inefficiencies and identified cost-saving opportunities across fulfillment and delivery workflows
- Flagged suspicious behaviors and helped improve fraud detection systems for voucher and discount abuse
- Assisted with voucher planning and budget allocation for targeted promotional campaigns

HONORS AND AWARDS

University Fellowship, University of California – Irvine

Merit-based, multi-year funding awarded for academic excellence and research potential

2023–2028

UC Irvine Graduate Dean's Recruitment Fellowship

2023

Bronze Medal, *Bronze Medallist in 12th Iranian National Olympiad in Astronomy and Astrophysics*

2016

Issued by Young Scholars Club (NODET)

Admitted to National Organization for Development of Exceptional Talents (NODET) School

2010

INTERESTS

Optimization, Machine Learning, Algorithmic Decision-Making, Network Optimization, Data Science

SKILLS AND SOFTWARES

Programming	Python; SQL
Data Analysis	A/B Testing; Experimental Data Analysis; Statistical Inference; SPSS
Modeling	Regression; Classification; Clustering
Optimization	Gurobi; CPLEX; AMPL; Integer Programming
Visualization	Tableau; Power BI; Matplotlib; Seaborn

SELECTED PROJECTS

- **Traffic Accident Severity Prediction** *ML & Data Analysis*
Built a classification pipeline using Python and scikit-learn to predict severity of U.S. traffic accidents. Compared multiple models including *Multinomial Logistic Regression*, *Decision Tree*, *Random Forest*, *XGBoost*, *KNN*, *MLP*, and *Neural Networks*. Preprocessed 100k+ records and evaluated with F1-score and confusion matrix.
- **Influencer Optimization via Network Centrality** *Network Analysis*
Identified key influencers in social graphs using *degree* and *closeness centrality*. Modeled influence spread using custom heuristics and simulated budget-constrained seeding strategies.
- **Churn Prediction and Segmentation** *Marketing Analytics*
Performed customer lifecycle analysis using *RFM clustering* and *logistic regression* to identify churn-prone segments. Built interpretable models and visualized feature impacts to guide retention strategies.
- **Airbnb Price Prediction** *Data Mining*
Engineered features from scraped Airbnb listings and applied regression models including *Linear*, *Ridge*, and *Decision Tree* regressors. Optimized performance using *GridSearchCV* and evaluated with RMSE.
- **Referral Program Simulation** *Marketing Strategy*
Developed an *agent-based simulation* of referral cascades and optimized reward policies under fixed budget. Measured impact on acquisition growth and campaign efficiency.
- **Truck Scheduling Model** *Transportation Optimization*
Improved an existing *MILP model* for truck sequencing in a cross-dock facility. Incorporated capacity and precedence constraints; solved with *Gurobi* to minimize total delay.
- **Multi-Stage Factory Simulation** *Modeling & Simulation*
Built a *discrete-event simulation* of a multi-stage production line to identify throughput bottlenecks. Evaluated performance under different input policies and resource allocations.

SELECTED COURSES

- **Data Science & Machine Learning:**
Machine Learning, Probabilistic Learning, AI and ML Business Research, Data Mining and its Applications, Regression Analysis, Statistical Quality Control, Analysis of Experimental Data
- **Optimization & Operations Research:**
Large-Scale Optimization, Operations Research, Queueing Theory and Probability Models, Dynamic Programming, Production and Inventory Control
- **Networks & Game Theory:**
Network Modeling and Applications, Game Theory
- **Systems & Business Management:**
Management Information Systems, Computer Information Systems, Project Control, Engineering Economics, Transportation Planning, Principles of Marketing, Business Economics, Microeconomics

VOLUNTEERING ACTIVITIES

Gamein 2020, Sharif University of Technology Vice President	Mar. 2020 – Nov. 2020
Scientific Association, Sharif University of Technology Member of the head council	Oct. 2019 - June. 2020
Gamein 2019, Sharif University of Technology Member of Marketing and Branding Team	Apr. 2019- Sep. 2019
Gamein 2018, Sharif University of Technology Member of Executive Team	May. 2018- Sep. 2018